No lecture Tuesday 4 March.
Next lecture Wednesday 5 March

Decay Width

Free particle hamiltonian $H = H_0$

$$H\psi = E\psi \longrightarrow \psi(x,t) = \psi(x)\, e^{-iE_j^0 t}$$

Time dependent Interaction $H = H_0 + V$

E_j: eigenvalues of energy.

$$E_j = E_j^0 + \langle j|V|j\rangle + \sum_{j\neq k} \frac{|\langle k|V|j\rangle|^2}{E_k - E_j} +$$

$$- i\pi \sum_{j\neq k} |\langle k|V|j\rangle|^2\, \delta(E_k - E_j) + O(3)$$

$$\Gamma = 2\pi \sum_{j\neq k} |\langle k|V|j\rangle|^2\, \delta(E_k - E_j)$$

Fermi Golden Rule: $\Gamma = 2\pi\, |M_{fi}|^2\, \rho(E)$

$1 \to 2 + 3 + \dots + n$

Two body Decay

$a \longrightarrow b + c$ $\quad S = 1$

$$\Gamma = \frac{1}{2m_a} \int |M|^2 \cancel{(2\pi)^4}\, \delta^4(p_a - p_b - p_c) \prod_{j=b,c} (2\pi)\, \boxed{\delta(p_j^2 - m_j^2)\, \theta(E_j)}\, \frac{d^4 p_j}{(2\pi)^4}$$

$$p_j^2 = E_j^2 - \vec{p}_j^{\,2} \qquad p_j^2 - m_j^2 = E_j^2 - \vec{p}_j^{\,2} - m_j^2$$

$$= E_j^2 - (\vec{p}_j^{\,2} + m_j^2)$$

$$\delta(f(x)) = \sum_i \frac{\delta(x - x_i)}{|f'(x_i)|} \qquad f(x_i) = \phi.$$

$$E_j = \pm\sqrt{\vec{p}_j^{\,2} + m_j^2}$$

$$\frac{\delta(E_j - E)}{2\sqrt{p_j^2 + m_j^2}}$$

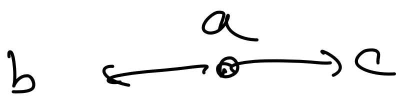
$$f(x) = E^2 = p^2 + m^2$$

$$2E = f'(x)$$

$$\Gamma = \frac{(2\pi)^{6-8}}{2m_a} \int |M|^2 \delta^4(\underline{p_a} - \underline{p_b} - \underline{p_c}) \frac{\delta(E - E_b)}{2\sqrt{p_b^2 + m_b^2}} \frac{\delta(E - E_c)}{2\sqrt{m_c^2 + p_c^2}} d^4\vec{p}_b \, d^4\vec{p}_c$$

$$d^4\vec{p}_j = dE_j \, d^3\vec{p}_j$$

$$= \frac{1}{2m_a} \frac{1}{4\pi^2} \frac{1}{4} \int |M|^2 \delta^4(\ldots) \frac{1}{\sqrt{p_b^2 + m_b^2}} \frac{1}{\sqrt{p_c^2 + m_c^2}} d^3\vec{p}_b \, d^3\vec{p}_c$$



$$\delta^4(\underline{p_a} - \underline{p_b} - \underline{p_c}) = \delta(E_a - E_b - E_c) \, \delta(\vec{p}_a - \vec{p}_b - \vec{p}_c)$$

rest frame of a: $\vec{p}_a = 0$. $E_a = m_a$

$$\delta(\vec{p}_a - \vec{p}_b - \vec{p}_c) \longrightarrow \delta(\vec{p}_b + \vec{p}_c) \Rightarrow \vec{p}_b = -\vec{p}_c$$

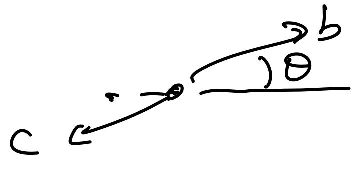
$$\vec{p}_b = -\vec{p}_c = \vec{p}^*$$

$$\delta(E_a - E_b - E_c) = \delta(m_a - E_b - E_c) \qquad m_a = E_b + E_c$$

$$\Gamma = \frac{1}{32\pi^2} \frac{1}{m_a} \int |M|^2 \delta(m_a - \underbrace{E_b - E_c}_{u}) \frac{1}{\sqrt{p^{*2} + m_b^2}\sqrt{p^{*2} + m_c^2}} d^3\vec{p}^*$$

$$p^* = \frac{m_a^2 - m_b^2 - m_c^2}{2m_a}$$

$$d^3p^* = p^{*2} \, dp^* \, d\Omega \qquad d\Omega = \sin\theta \, d\theta \, d\varphi$$



$$\delta(m_a - E_b - E_c)$$

$$u = E_b + E_c = \sqrt{p^2 + m_b^2} + \sqrt{p^2 + m_c^2}$$

$$\frac{du}{dp} = \frac{2p}{2\sqrt{p^2 + m_b^2}} + \frac{2p}{2\sqrt{p^2 + m_c^2}} = \frac{\cancel{2}p\left(\overbrace{\sqrt{p^2 + m_b^2} + \sqrt{p^2 + m_c^2}}^{u}\right)}{\cancel{2}\sqrt{p^2 + m_b^2}\,\sqrt{p^2 + m_c^2}}$$

$$= \frac{pu}{\sqrt{}\;\sqrt{}}$$

$$\frac{p}{\sqrt{p^2 + m_b^2}\,\sqrt{p^2 + m_c^2}} = \frac{1}{u}\,\frac{du}{dp}$$

$$\frac{1}{\sqrt{p^2 + m_b^2}\,\sqrt{p^2 + m_c^2}} \quad \overbrace{p^2\,dp\;\,d\Omega}^{d^3p}$$

$$\delta(m_a - u) = \frac{1}{u}\,\frac{du}{dp}\;p\,dp\,d\Omega.$$

$$= \frac{p}{m_a}\,d\Omega$$

$$\Gamma = \frac{1}{32\pi^2}\,\frac{1}{m_a}\int |M|^2\,\frac{p}{m_a}\,d\Omega.$$

$$M = \langle f | H_I | i \rangle$$

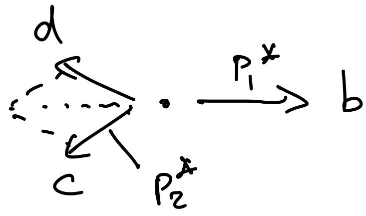
$$= \frac{1}{32\pi^2}\,\frac{1}{m_a^2}\;p\int |M|^2\,d\Omega.$$

$$b \longleftarrow \overset{a}{\circ} \longrightarrow c \qquad p = \frac{m_a^2 - m_b^2 - m_c^2}{2m_a}.$$

$$Q = m_a - m_b - m_c$$

phase space

Exercise: $a \rightarrow b + c + d$



$\Gamma = \frac{1}{\tau}$ $\quad\quad$ $\underbrace{T = L}_{c=1} \underbrace{= E^{-1}}_{\hbar=1}$

$[\Gamma] = E$. $\quad\quad$ τ: mean life time

$[m, p] = E$ $\quad\quad$ Γ: total decay width

$[|M|] = E$

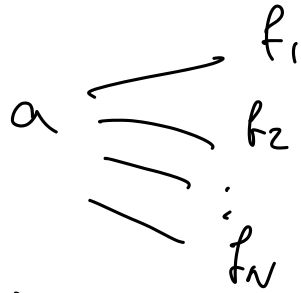
$\pi^+ \to \mu^+ \nu_\mu$

$\quad \to e^+ \nu_e$

$Z^0 \to e^+ e^-$
$\mu^+ \mu^-$
$\tau^+ \tau^-$
$\nu_e \bar{\nu}_e$
$\nu_\mu \bar{\nu}_\mu$
$\nu_\tau \bar{\nu}_\tau$
$q \bar{q}$

$\Gamma(Z^0 \to e^+ e^-)$

$\Gamma(Z^0 \to \mu^+ \mu^-)$



$$\Gamma_{tot} = \sum_j^N \Gamma(a \to f_j)$$

$$BF(a \to f_j) = \frac{\Gamma(a \to f_j)}{\Gamma_{tot}} = [0, 1]$$

$$H \to \gamma\gamma$$
$$Z^0 Z^0$$
$$W^+ W^-$$
$$b\bar{b}$$

$$\pi^+ \to \mu^+ \nu_\mu$$
$$\to e^+ \nu_e$$

π : $m = 140$ MeV

μ : 106 MeV.

e : 0.511 MeV

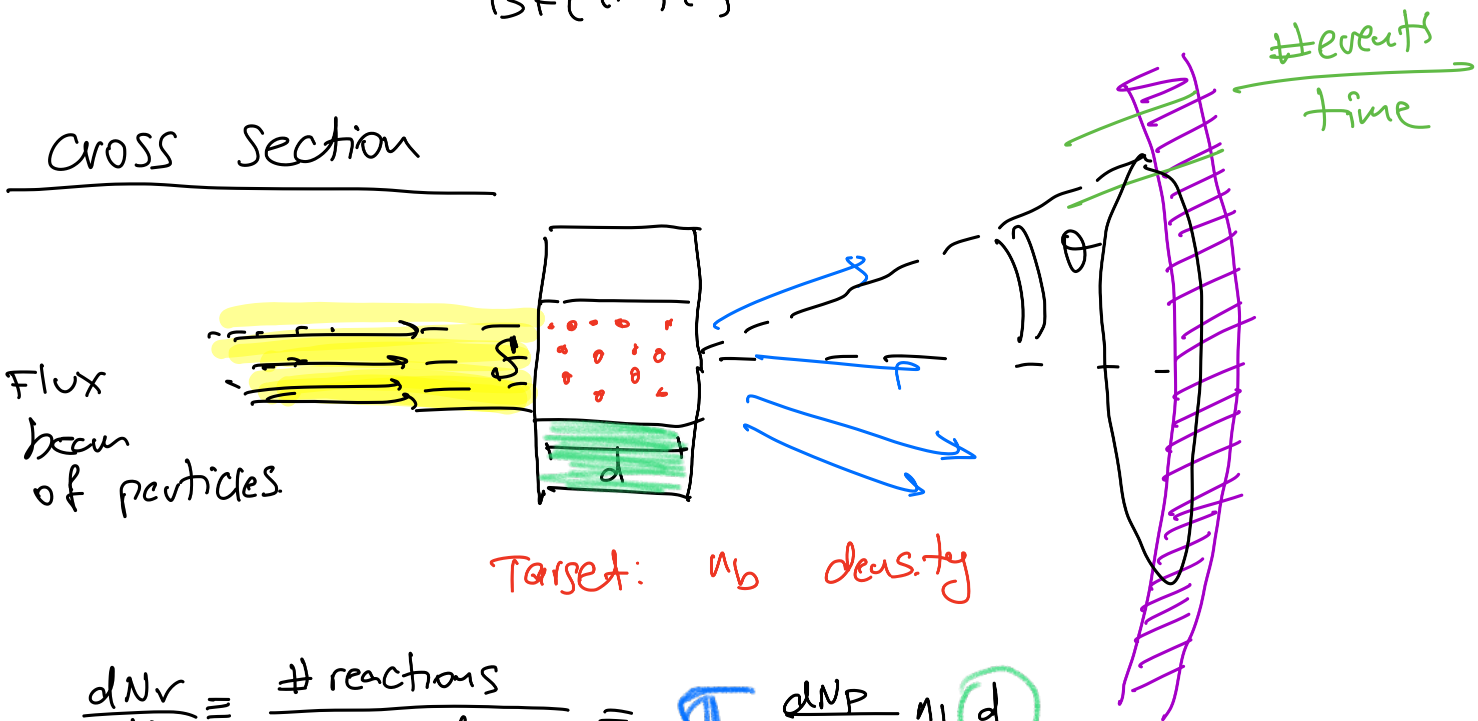
$$Q = m_\pi - m_e \qquad Q_{\pi\to\mu} = 34 \text{ MeV}$$
$$Q_{\pi\to e} = 140 \text{ MeV}$$

phase space $\Rightarrow$ e preferred over μ.

$$Q_e >> Q_\mu.$$

In reality $\dfrac{BF(\pi\to\mu)}{BF(\pi\to e)} \simeq 10^4$

Cross Section



$$\frac{dN_r}{dt} = \frac{\#\text{ reactions}}{\text{time of observation}} = \sigma \frac{dN_p}{dt} n_b d \to \text{target thickness}$$

$$T^{-1} \qquad L^2 \qquad T^{-1} L^{-3} L$$

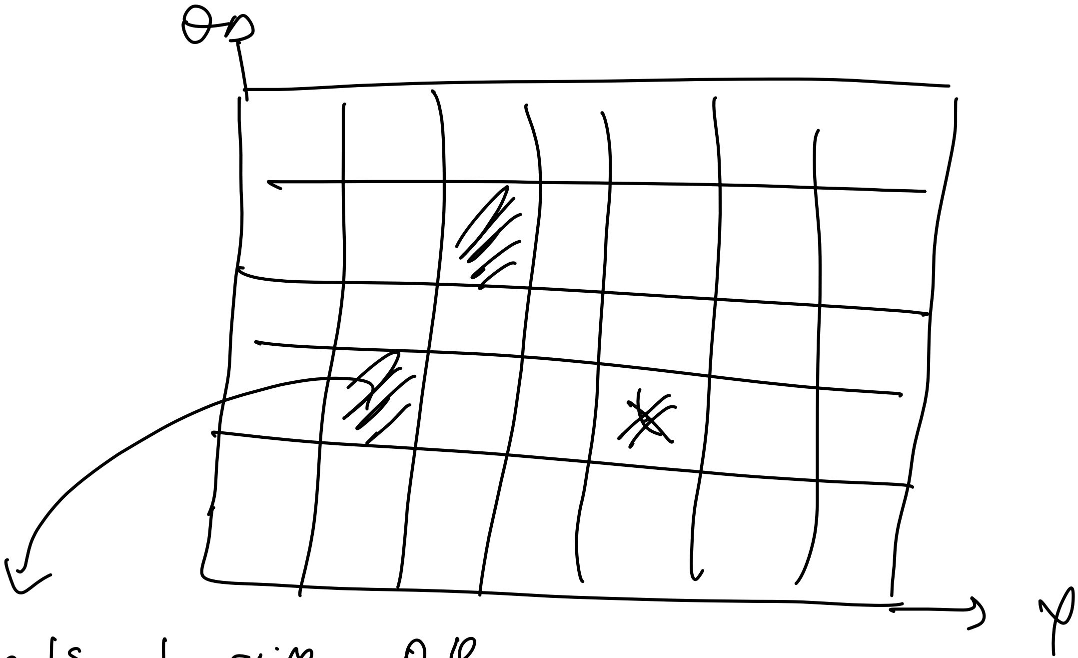
σ: cross section

Differential cross section: $d\Omega = \sin\theta \, d\theta \, d\varphi$

events as function of something.

$\frac{d\sigma}{d\Omega}$

$\frac{d\sigma}{d\theta}$

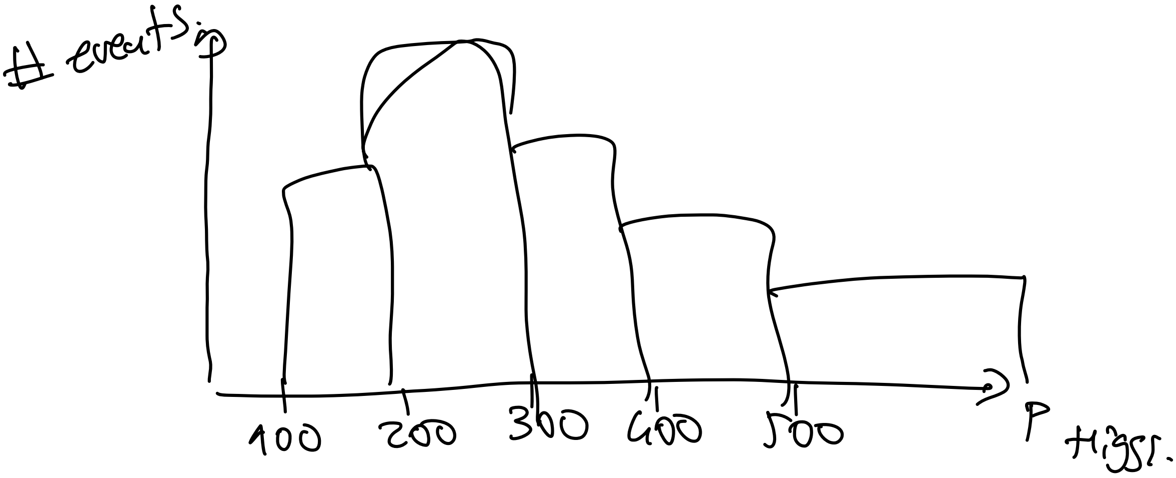


events at given θ, φ

$$\frac{d\sigma}{d\Omega} = \frac{\#\text{ events}}{\Delta\varphi \, \Delta\theta}$$

$$\frac{d^3\sigma}{dE\,d\theta\,d\varphi}$$

$\sigma(p+p \to H+X)$: total Higgs xsection



$\frac{d\sigma}{dP_{Higgs}}$

$$\frac{dN_r}{dt} = \sigma \frac{dN_p}{dt} n_b \cdot d = \sigma \underbrace{\frac{dN_p}{dt}\frac{1}{S}}_{\Phi_p} n_b \cdot d \cdot S$$

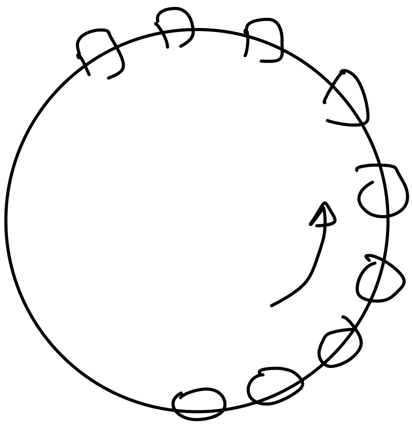
$$= \sigma \, \Phi_p \, N_B$$

$$\frac{1}{N_B} \frac{dN_r}{dt} = \sigma \cdot \Phi_p$$

N_B : # bunches. 3800.

LHC:

27 km



every bunch: 10^{11} protons.

$$\frac{dN_r}{dt} = \sigma \, N_B \Phi_p = \sigma \, N_B \underbrace{\frac{N_p}{S} \gamma f_{rev}}_{\text{depend on accelerator}}$$

$$\frac{dN_r}{dt} = \sigma \cdot \mathcal{L}_{inst}$$ instantaneous luminosity

$[\sigma] = cm^2$ $\quad [\mathcal{L}] = cm^{-2} s^{-1}$

$$N_{reactions} = \sigma \cdot \mathcal{L}_{inst} \cdot \Delta t$$

σ → depends on the process.

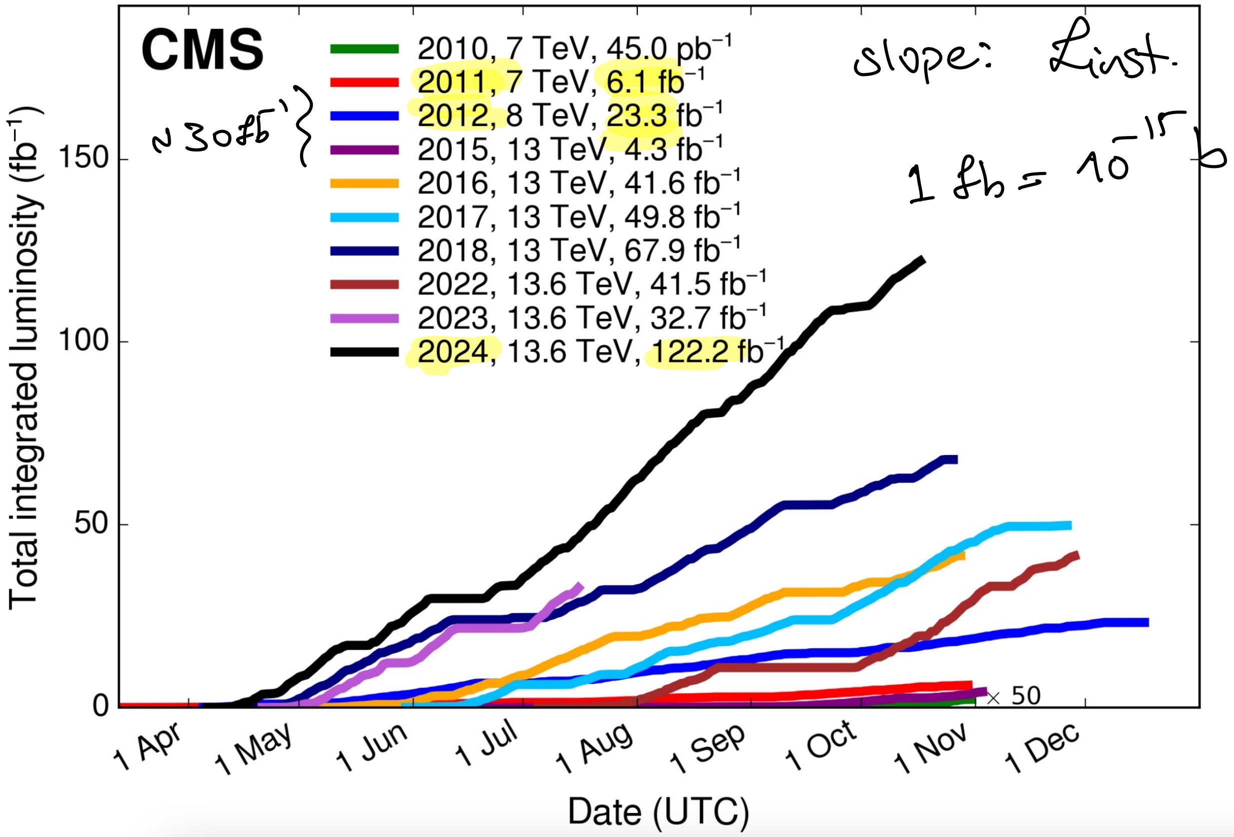
Δt → time of data taking.

assumes same $\mathcal{L}_{inst}$ during Δt

$$1 \, b = 10^{-24} \, cm^2 = 10^{-28} \, m^2$$

$$\mathcal{L}_{int} = \mathcal{L}_{inst} \cdot \Delta t = \int \mathcal{L}_{inst} \cdot dt$$

integrated luminosity

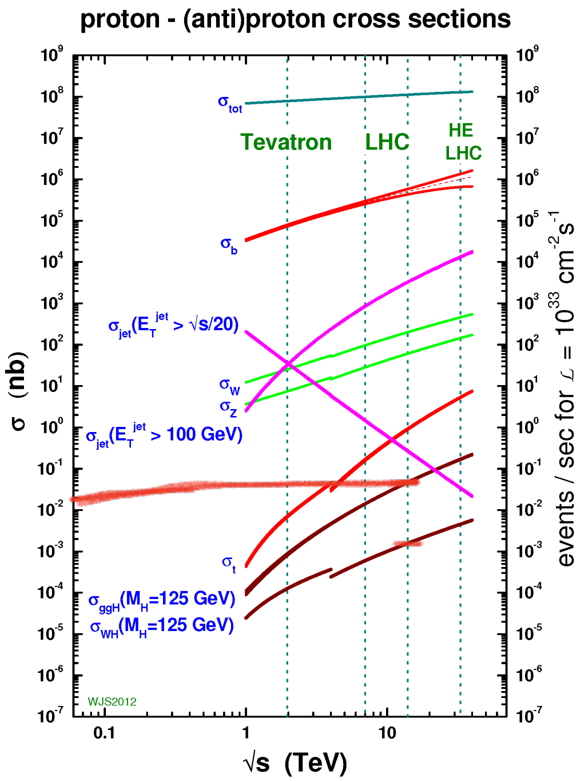


$\sigma(p+p \to H+X) = 10^{-2}$ nb

$L_{int} = \int L_{inst}\, dt = 100\ nb^{-1}$

$\#\text{events} = \sigma \cdot L_{int} = 100 \cdot 10^{-2} = 1.$

1 Higgs every $100\ nb^{-1}$



$L_{2024} = 100\ fb^{-1}$

$1\ fb = 10^{-6}\ nb$

$1\ fb^{-1} = 10^{6}\ nb^{-1}$

$N^{\text{in } 2024}_{\text{Higgs}} = 10^{-2}\ nb \times 100 \times 10^{6}\ nb^{-1}$

$= 10^{6}$ Higgs.

$H \to \gamma\gamma$

$$\underset{2024}{N}(H \to \gamma\gamma) = \sigma_{H.} \times L_{int}^{2024} \times BF(H \to \gamma\gamma)$$

$$= 10^{6} \times 10^{-4} = 100$$

$$H \to Z^0 \; Z^0$$

$Z^0 \hookrightarrow \mu^+\mu^-$

$Z^0 \hookrightarrow e^+e^-$

$p+p \to H + X$

$H \hookrightarrow Z \; Z$

$Z \hookrightarrow \mu\mu$

$Z \hookrightarrow e^+e^-$

$$N(H \to e^+e^-\mu^+\mu^-) = 10^{6}_{H.} \times BF(H \to ZZ) \times$$

$$\times BF(Z \to e^+e^-) \quad 3\%$$

$$\times BF(Z \to \mu^+\mu^-)$$

$$(3 \times 10^{-2})^{3}$$

$$30 \times 10^{-6} = 3 \times 10^{-5}$$

$$10^{6} \times 3 \times 10^{-5} = 30$$

Higgs decays at m_H=125GeV